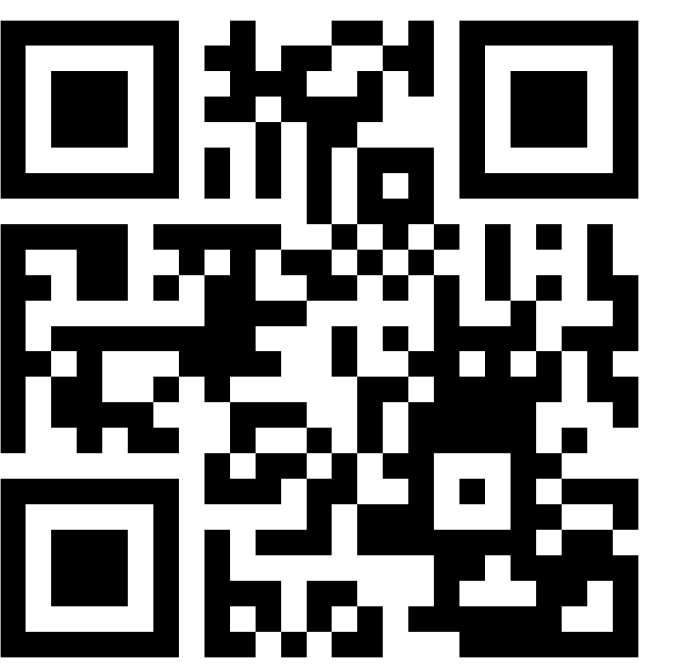




INTRODUCTION

Polygenic risk scores (PRS) estimate inherited disease risk from genome-wide association study (GWAS)-identified variants, offering a route to precision prevention. Type 2 diabetes (T2D) is projected to affect 783 million people by 2045, yet 91% of GWAS participants are of European (EUR) ancestry.^[1,2] Central and South Asian (CSA) populations carry the highest T2D burden but remain understudied, and it is unclear whether poor PRS transferability reflects divergent biology, technical artifacts (mismatched linkage disequilibrium (LD) and allele frequencies), or discovery power, which are factors explaining 70-80% of cross-ancestry accuracy loss.^[3,4]

Hypothesis: Limited CSA discovery power, is the dominant driver of poor cross-ancestry T2D PRS transferability.

METHODS

EUR and CSA T2D (ICD-10: E11) summary statistics came from the Pan-UK Biobank; individual genotypes came from 1000 Genomes Phase 3 EUR and South Asian (SAS) panels.^[5,6] Quality control (MAF > 1%, missingness < 5%, HWE $p > 1 \times 10^{-6}$) and PCA-based ancestry verification left 503 EUR and 478 SAS individuals.^[7,8] Two PRS methods were compared: clumping-and-thresholding (C+T; LD $r^2 < 0.1$) and LDpred2-inf, which jointly models genome-wide LD.^[8,9] Four configurations isolated effect-size, LD-reference, and discovery-power contributions to transferability. Outcomes included mean score shift, top-decile enrichment, cross-score concordance (Pearson r), locus-level heterogeneity (Cochran's Q, Bonferroni-corrected), and genome-wide genetic correlation (rg).^[10,11]

Miscalibration

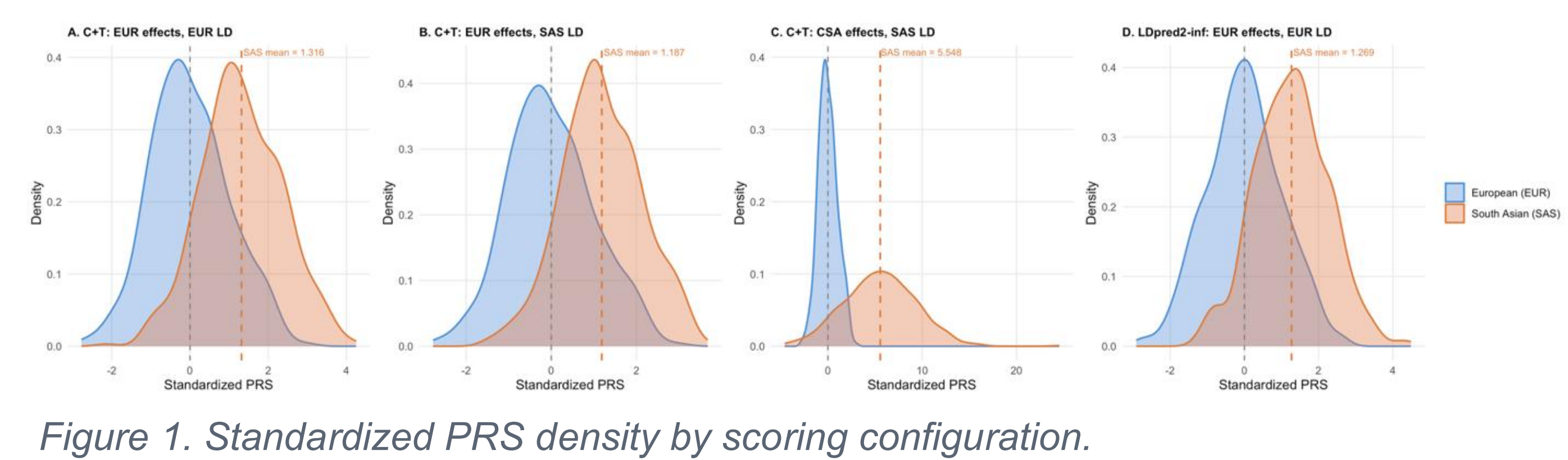


Figure 1. Standardized PRS density by scoring configuration.

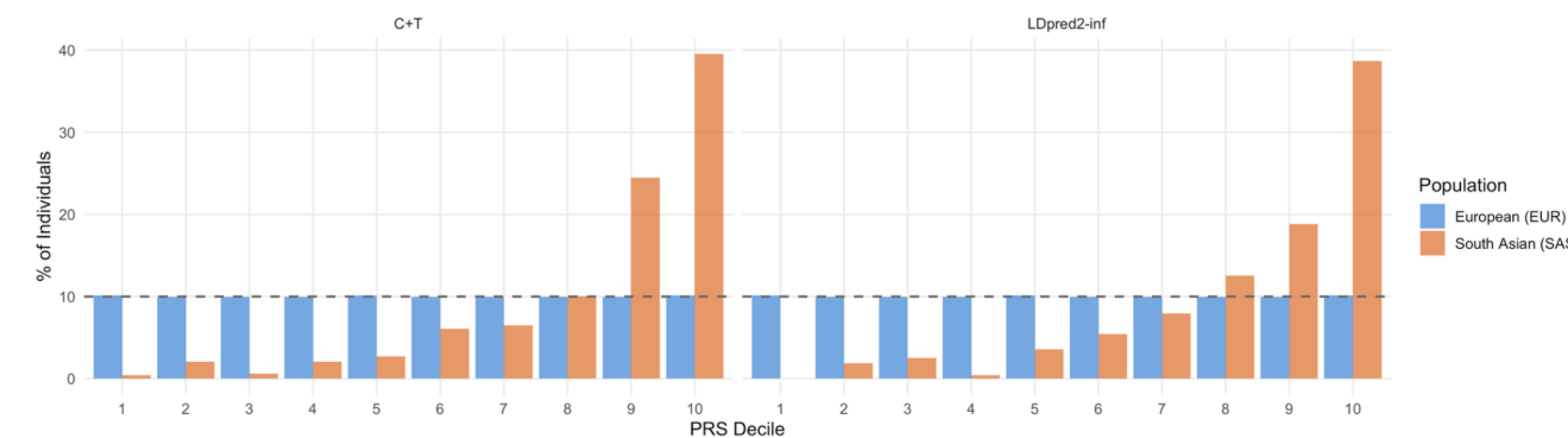


Figure 2. PRS decile distribution. Dashed line = 10% expected.

EUR-derived PRS shifted SAS scores 1.32 SD (C+T) / 1.27 SD (LDpred2-inf) above EUR (Fig. 1), producing ~4-fold enrichment of SAS individuals in the EUR-defined top risk decile (Fig. 2).

Discovery Power

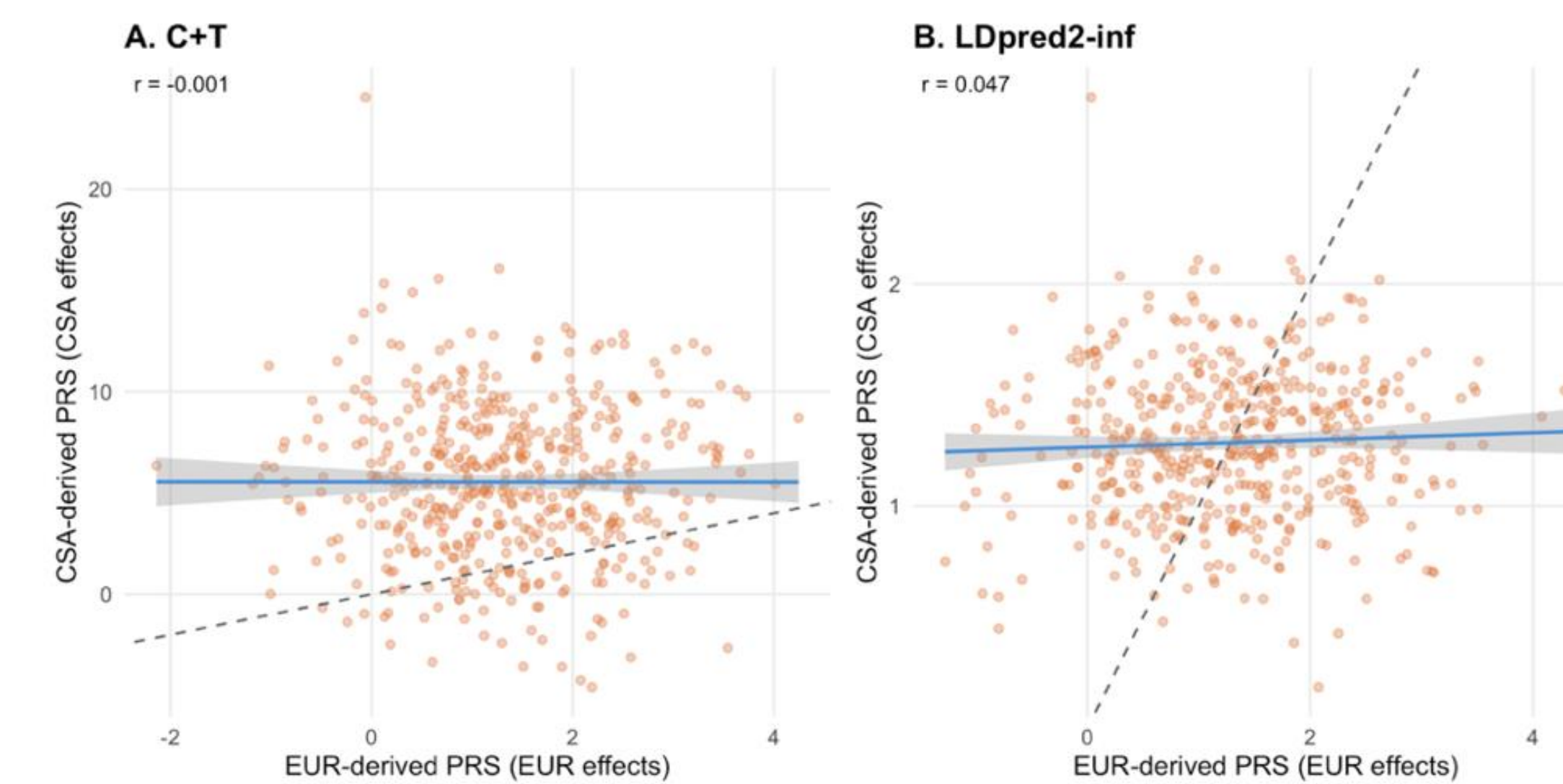


Figure 3. EUR- vs. CSA-derived score concordance within 478 SAS individuals (C+T, left; LDpred2-inf, right). Dashed line = perfect concordance.

Cross-score concordance was negligible ($r = -0.001$ to 0.047), and directional agreement rose from 63.7% at low CSA $|Z|$ to 100% at high $|Z|$.

Table 1. Directional concordance of effect estimates

CSA $ Z $ bin	N SNPs	% same direction as EUR
< 1	5,806	63.7
1–2	2,342	71.3
2–3	439	98.0
> 3	202	100

Note. 0 of 8,789 loci showed Bonferroni-significant EUR–CSA heterogeneity (Cochran's Q).

Technical Artifacts

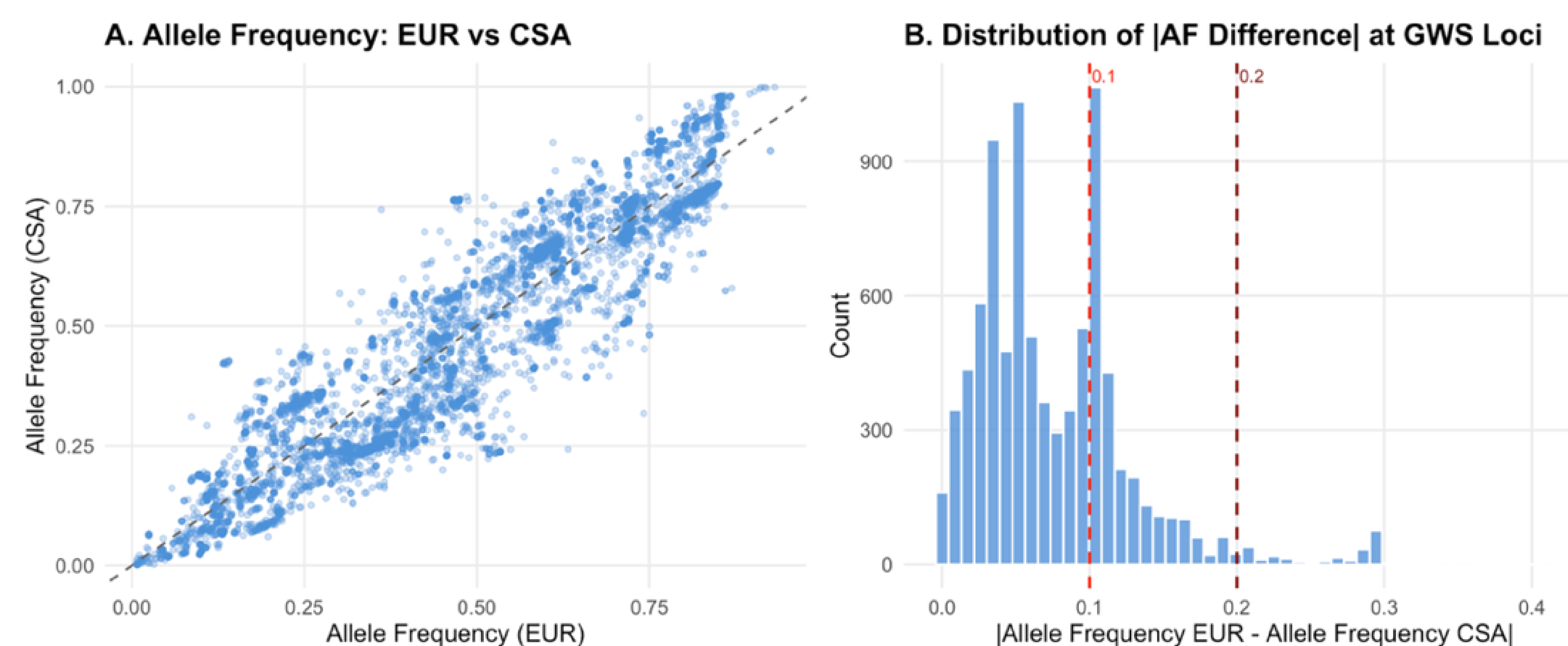


Figure 4. (A) Risk allele frequency, EUR vs. CSA, at 8,789 EUR GWAS T2D loci. (B) Distribution of absolute allele-frequency differences.

Most loci had similar allele frequencies (mean $\Delta = 0.076$; divergent subset 0.25–0.30); ancestry-matched LD reduced the C+T shift to 1.19 SD from 1.32.

Effect Size

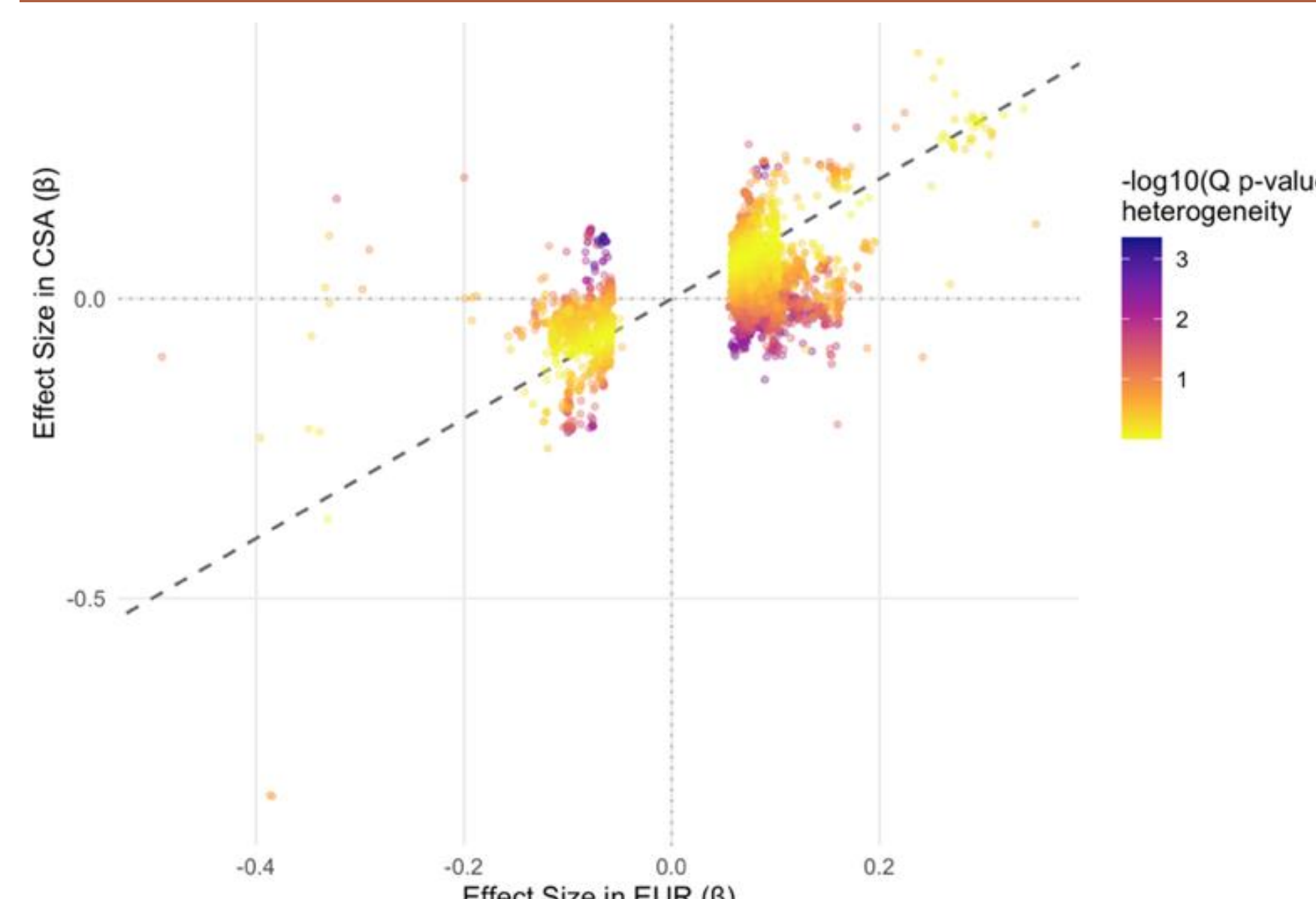


Figure 5. EUR vs. CSA effect size at 8,789 EUR genome-wide-significant loci, colored by heterogeneity (Cochran's Q). Points cluster on the diagonal at well-powered loci.

Genome-wide genetic correlation between EUR and CSA was high ($rg = 1.27$), and locus-level effect sizes were broadly concordant ($r = 0.417$ overall; $r = 0.609$ restricted to $|Z| \geq 1$), though CSA estimates were systematically attenuated relative to EUR. No locus, including TCF7L2, showed significant heterogeneity after Bonferroni correction (0/8,789 loci).

DISCUSSION and CONCLUSION

Interpretation: EUR-derived PRS showed miscalibration in South Asians. The entire score distribution shifted (Fig. 1A & 1D), and nearly four times too many South Asians were flagged high-risk (Fig. 2). Divergent biology cannot explain this, since rg was 1.27, indicating essentially shared genetic architecture and 0/8,789 loci showed real heterogeneity (Fig. 5, Table 1).^[10] The scoring algorithm was not responsible either, since LDpred2-inf models linkage far more carefully than clumping and thresholding,^[8,9] yet it transferred no better (Fig. 3). Technical artifacts played some role, as allele frequencies diverged at a subset of loci (Fig. 4) and swapping in an ancestry-matched LD reference only partially narrowed the gap (Fig. 1B vs. 1A).

Limitation(s): The 1000 Genomes panel has no phenotype data, so transferability here was assessed through genotype-level metrics rather than predictive accuracy in diagnosed patients.

Future Work: Expanding South Asian-specific genetic studies, rather than building a smarter model, is therefore the most direct path to equitable diabetes risk prediction, echoing calls to broaden ancestral representation before PRS reach the clinic.^[12] Future work can include expanding CSA-focused GWAS efforts (e.g., DIAMANTE, SAT2D, South Asia Biobank) using phenotype cohorts to validate PRS against clinical outcomes.

Conclusion: Discovery power is the dominant driver of poor transferability and technical artifacts explain only part of the remaining gap.

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